Co-funded by
the European Union

Erasmus+ project ROBO STEAM ACADEMY
KA220-VET-7CF4F308

Lap Timer Bill of Materials and Print Settings

Version 2 parts, files and prototype settings · Version 2.0

Table of Contents

About this guide....1

1. Main hardware....1

2. Printable files....2

3. Version 2 CAD values....2

4. PCB dimensions used by the model....3

5. Starting slicer settings....3

5.1 Orientation....3

6. Tools for assembly and inspection....3

7. Procurement and classroom planning....4

About this document....5

About this guide

This document separates confirmed project items from version 2 starting values. Quantities cover one enclosure. Electrical variants and supplier part numbers must be confirmed before purchasing for a class set.

1. Main hardware

Item	Quantity	Specification	Status
Controller module	1	NodeMCU V3 ESP8266	Project requirement
Expansion baseboard	1	HW-389-style NodeMCU Base Ver 1.0	Project requirement
Reflective sensor	1	E18-D80NK, M18 x 1 mounting thread	Project requirement
Printed base	1	`base_v2.stl`	Version 2 prototype
Printed top	1	`top_v2.stl`	Version 2 prototype
Closure screws	2	3 mm, socket-head or pan-head	Type and length to verify
Sensor retaining nuts	As supplied	Compatible with actual sensor	Verify physical set
Disconnectable sensor joint	1	Three-wire connection	Exact part to verify
USB cable	1	Micro-USB with molded plug body	Measure required cable

Item	Quantity	Specification	Status
Power adapter	1	Project-approved output and connector	Not yet specified

Safety

Do not buy or connect a mains adapter from this table alone. The verified voltage, polarity, current rating, and safety requirements are still missing.

2. Printable files

File	Quantity	Purpose
`pcb_fit_coupon_v2.stl`	1	Check the scan-derived hole pattern and PCB clearances
`sensor_nut_check_v2.stl`	1 or more	Confirm the M18 panel-hole fit on this printer
`screw_coupon_v2.stl`	1 or more	Test nearby pilot sizes and screw type
`base_v2.stl`	1	Hold the PCB and lower connector passages
`top_v2.stl`	1	Hold the sensor and close the enclosure

3. Version 2 CAD values

Parameter	Value	Status
Main shell outside X	69.1 mm	CAD nominal
Main shell outside Y	70.3 mm	CAD nominal
Overall assembled Z	61.0 mm	CAD nominal
Bottom tray seam Z	11.0 mm	Prototype value
Sensor axis Z	45.0 mm	User-selected version 2 height
Sensor panel hole	19.0 mm	Passed version 1 fit test
Wall thickness	2.4 mm	Starting value
Floor thickness	2.4 mm	Starting value
Roof thickness	2.0 mm	Starting value
Mating clearance	0.30 mm per side	Starting value
PCB rear clearance	1.0 mm	User-selected connector-edge clearance
PCB left and right clearance	3.0 mm per side	Clone-board allowance
PCB front clearance	3.0 mm	Clone-board allowance
PCB locating peg	2.8 mm with tapered lead-in	Must be fit-tested
Screw pilot	2.6 mm centre coupon value	Must be tested
Top lug clearance	3.4 mm	Prototype value
Screw-head recess	6.5 mm	Prototype value

4. PCB dimensions used by the model

The PCB coordinate origin is its rear-left corner when viewed from above. X runs from rear to front. Y runs from left to right.

Item	Value	Evidence
PCB X length	60.3 mm	Complete 600 dpi physical-board scan, +/-0.3 mm
PCB Y width	59.5 mm	Complete 600 dpi physical-board scan, +/-0.3 mm
PCB thickness	1.6 mm	Existing project reference; verify physical board
Hole 1	X 4.6, Y 14.6 mm	Physical-board scan, +/-0.2 mm per axis
Hole 2	X 4.6, Y 39.6 mm	Physical-board scan, +/-0.2 mm per axis
Hole 3	X 56.9, Y 14.6 mm	Physical-board scan, +/-0.2 mm per axis
Hole 4	X 56.9, Y 39.6 mm	Physical-board scan, +/-0.2 mm per axis

The scan suggests an approximately 3.4 mm hole opening. The blurred plated edge is not precise enough to approve that value without a physical fit test.

5. Starting slicer settings

Setting	Version 2 starting value	Record after print
Process	FDM	
Nozzle	0.4 mm	
Layer height	0.2 mm	
Material	PLA or PETG	
Supports	Avoid where practical	
Wall count	Set by operator	
Top and bottom layers	Set by operator	
Infill pattern and percentage	Set by operator	
Temperatures	Follow material and printer guidance	
Build-plate adhesion	Set by operator	

5.1 Orientation

Print the base with its floor on the build plate. The supplied top STL is oriented with its roof on the build plate and the locating lip upward. Print both coupons on their largest flat face.

Check: the slicer preview shows no part floating above the build plate.

6. Tools for assembly and inspection

Tool	Purpose
Digital calipers	Measure printed parts and hardware interfaces
Suitable screwdriver	Install the selected closure screws
Deburring tool or craft knife	Remove loose brim and strings
Eye protection	Protect during cleanup and fit testing
Camera or phone	Record fit problems and version 3 changes

Safety

Cut away from hands and cables. A teacher should supervise sharp-tool use according to workshop rules.

7. Procurement and classroom planning

Buy one physical sample of each electronics variant before ordering a class set. Fit it to the version 2 print, record the measurements, and update the files. A matching product title does not prove that clone boards have matching holes, connectors, or height.

About this document



Co-funded by
the European Union

















RoboSTEAM Lap Timer — a result of the Erasmus+ project **ROBO STEAM ACADEMY** (KA220-VET — Cooperation partnerships in vocational education and training, project number KA220-VET-7CF4F308, 2024–2027), used for the project's robot competitions (Work Package 3). Project website: https://robosteam.eu/

Partners: National STEM Center (coordinator, Bulgaria, https://stem.mon.bg/) · Institute of Information and Communication Technologies, Bulgarian Academy of Sciences (Bulgaria, https://www.iict.bas.bg/) · Academy Nikola Tesla (Bulgaria, https://ant.bg/) · Technical University of Košice (Slovakia, https://www.tuke.sk/) · Stredná priemyselná škola elektrotechnická Prešov (Slovakia, https://spseadlerka.edupage.org/) · National Association of Public Librarians and Libraries in Romania – ANBPR (Romania, https://anbpr.org.ro/) · Liceul Tehnologic Cezar Nicolau (Romania, https://www.cezar-nicolau.ro/).

Licence: texts, images and 3D-print files **CC BY 4.0** (Creative Commons Attribution 4.0 International, https://creativecommons.org/licenses/by/4.0/); program code **MIT licence**. You may share and adapt the materials, also commercially, if you credit "ROBO STEAM ACADEMY, Erasmus+ project KA220-VET-7CF4F308" and indicate changes. Third-party images, code and parts keep their own licences and are credited where they are used.

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the Human Resource Development Centre (HRDC). Neither the European Union nor the granting authority can be held responsible for them.